Lesson 4.4 - How to Make Box and Whisker Plots

> **Video:** How To Make Box and Whisker Plots by The Organic Chemistry Tutor **Chapter:** 4 - Measures of Position

The five-number summary

A box-and-whisker plot (box plot) is built from five values:

1. **Minimum** (smallest value that is not an outlier)
2. Q_1
3. Q_2 (median)
4. Q_3
5. **Maximum** (largest value that is not an outlier)

Steps

1. Sort the data.
2. Find Q_2, then Q_1 and Q_3.
3. Compute $\mathrm{IQR} = Q_3 - Q_1$ and the fences $Q_1 - 1.5\,\mathrm{IQR}$ and $Q_3 + 1.5\,\mathrm{IQR}$.
4. Values outside the fences are **outliers**. They are plotted as **separate points**, not as part of the whiskers.
5. Draw a number line. Draw a **box** from Q_1 to Q_3 with a line at Q_2, and **whiskers** out to the minimum and maximum non-outliers.

Example 1 - no outliers

Data: 11, 22, 20, 14, 29, 8, 35, 27, 13, 49, 10, 24, 17

Sorted ($n = 13$): 8, 10, 11, 13, 14, 17, **20**, 22, 24, 27, 29, 35, 49

- $Q_2 = 20$
- Lower half 8, 10, 11, 13, 14, 17 → $Q_1 = \dfrac{11 + 13}{2} = 12$
- Upper half 22, 24, 27, 29, 35, 49 → $Q_3 = \dfrac{27 + 29}{2} = 28$
- $\mathrm{IQR} = 28 - 12 = 16$
- Fences: $12 - 1.5(16) = -12$ and $28 + 1.5(16) = 52$

Both 8 and 49 are inside $[-12, 52]$, so there are **no outliers**. Minimum = 8, maximum = 49.

```
           +-------|-------+
        |---|       |       |--------------------|
           +-------|-------+
+---------+---------+---------+---------+---------+
0         10        20        30        40        50
         min=8  Q1=12  Q2=20  Q3=28             max=49
```

Example 2 - with an outlier

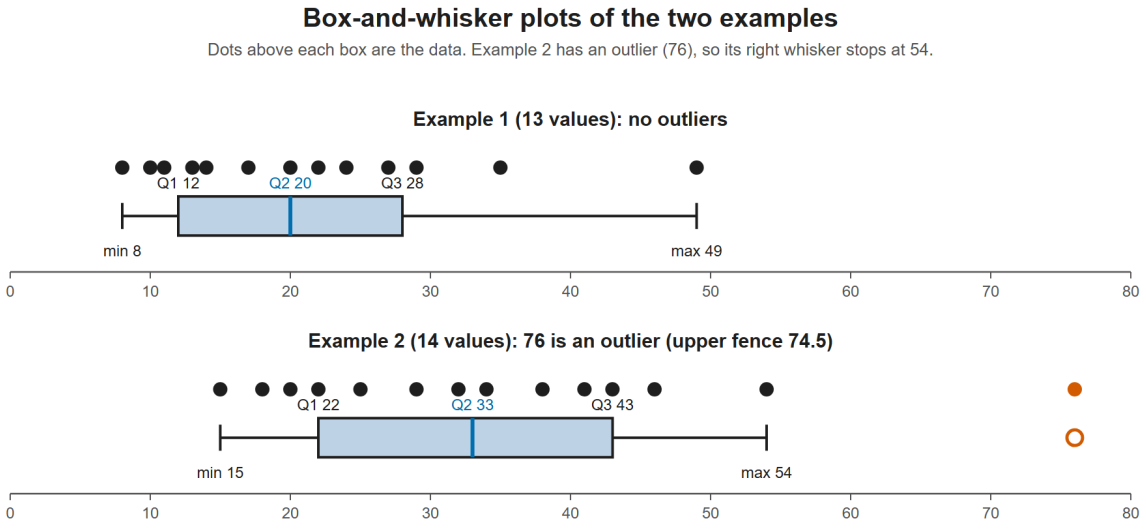
Data: 18, 34, 76, 29, 15, 41, 46, 25, 54, 38, 20, 32, 43, 22

Sorted ($n = 14$): 15, 18, 20, **22**, 25, 29, 32 || 34, 38, 41, **43**, 46, 54, 76

- $Q_2 = \dfrac{32 + 34}{2} = 33$
- Lower half (7 values) → middle value $Q_1 = 22$
- Upper half (7 values) → middle value $Q_3 = 43$
- $\text{IQR} = 43 - 22 = 21$
- Fences: $22 - 1.5(21) = -9.5$ and $43 + 1.5(21) = 74.5$

76 > 74.5, so 76 is an outlier. The right whisker stops at the largest non-outlier, **54**, and 76 is drawn as a separate point ($*$).

```
                  +----------|---------+
              |---|          |         |----------|                    *
                  +----------|---------+
+---------+---------+---------+---------+---------+---------+---------+---------+
0         10        20        30        40        50        60        70        80
            min=15  Q1=22    Q2=33    Q3=43     max=54              outlier=76
```



Both examples drawn to scale with the data above each box. In Example 2, 76 is beyond the upper fence of 74.5, so the right whisker stops at the largest non-outlier (54) and 76 is drawn as its own point.

Key takeaways

- Box = Q_1 to Q_3 (the IQR), with a line at the median Q_2.
- Whiskers reach the most extreme values that are **not** outliers.
- Outliers (beyond $Q_1 - 1.5\,\mathrm{IQR}$ or $Q_3 + 1.5\,\mathrm{IQR}$) are shown as individual points.